with civic performance dashboards, city administrations have improved route scheduling, increased accountability among waste contractors, and strengthened environmental hygiene interventions under the national “Clean India” campaign.

Another impactful application is disaster risk assessment and flood resilience planning. For example, in Hyderabad and Chennai, UAVs are used to monitor lake boundaries, detect water encroachments, and model drainage flows to prevent urban flooding. During emergency situations such as the Kerala floods (2018), drones were deployed for search and rescue coordination, damage assessment, and relief distribution, helping authorities respond more rapidly and effectively. Similarly, in Varanasi, drone-based Ganga riverfront mapping supports both heritage conservation and water quality monitoring efforts.

The integration of UAVs under India’s Smart City Mission highlights a shift toward data-driven urban governance, where real-time spatial intelligence guides policymaking. While challenges remain including airspace regulatory constraints, operator training requirements, and cybersecurity safeguards the expanding role of drones demonstrates India’s commitment to leveraging affordable, scalable, and technologically advanced solutions to build more resilient, efficient, and citizen-responsive urban systems.

8.10.6 Case study: Zipline – medical logistics in Rwanda and Ghana

Zipline represents one of the most successful large-scale implementations of UAV technology in medical logistics. Launched in Rwanda in 2016 and later expanded to Ghana in 2019, Zipline’s drone delivery network was designed to solve the problem of delayed access to life-saving medical supplies in regions where geographic barriers, road conditions, and limited